Abstract Algebra | Quiz 6 | October 14, 2026

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1. Find the order of $(8,8,8)$ in $\mathbb{Z}_{10} \times \mathbb{Z}_{24} \times \mathbb{Z}_{80}$.

In $\mathbb{Z}_{10}$, $|8| = \frac{10}{\gcd(10,8)} = \frac{10}{2} = 5$

In $\mathbb{Z}_{24}$, $|8| = \frac{24}{\gcd(24,8)} = \frac{24}{8} = 3$

In $\mathbb{Z}_{80}$, $|8| = \frac{80}{\gcd(80,8)} = \frac{80}{8} = 10$

$\therefore |(8,8,8)| = \operatorname{lcm}(5,3,10) = \boxed{30}$

2. Prove or disprove: $U(10) \cong U(12)$.

$U(10) = \{1,3,7,9\}$

Note: $\langle 3 \rangle = \{1,3,9,7\} = U(10)$

so $\boxed{U(10) \text{ is cyclic}}$

$U(12) = \{1,5,7,11\}$

Note: $\langle 1 \rangle = \{1\} \neq U(12)$

$\langle 5 \rangle = \{1,5\} \neq U(12)$

$\langle 7 \rangle = \{1,7\} \neq U(12)$

$\langle 11 \rangle = \{1,11\} \neq U(12)$

Therefore $\boxed{U(12) \text{ is } \underline{\text{not}} \text{ cyclic}}$

Conclusion $U(10) \ncong U(12)$ because one is cyclic and the other is not.

3. Prove or disprove: $\mathbb{Z} \cong \mathbb{Q}$ (The operation in both groups is $+$.)

Notice that $\mathbb{Z} = \langle 1 \rangle$, so $\mathbb{Z}$ is cyclic, But $\mathbb{Q}$ is not cyclic because for any $\frac{p}{q} \in \mathbb{Q}$, $\langle \frac{p}{q} \rangle = \{ \frac{kp}{q} : k \in \mathbb{Z} \} \neq \mathbb{Q}$

Therefore $\mathbb{Z} \ncong \mathbb{Q}$ because one is cyclic and the other is not cyclic.

4. Prove or disprove: If G and H are groups, then $G \times H \cong H \times G$.

Proof Define $\varphi : G \times H \to H \times G$ as $\varphi((x,y)) = (y,x)$. This is clearly a bijection. Also observe that

$$\begin{aligned} \varphi((x,y)(u,v)) &= \varphi((xu, yv)) \\ &= (yv, xu) \\ &= (y\,x)(v\,u) \\ &= \varphi(x,y)\,\varphi(u\,v) \end{aligned}$$

Because $\varphi((x,y)(u,v)) = \varphi(x,y)\,\varphi(u,v)$ the bijection φ is an isomorphism.

Therefore $G \times H \cong H \times G$